Journal of the Korean Ceramic Society (2024) 61:161–169
https://doi.org/10.1007/s43207-023-00338-w

ORIGINAL ARTICLE

Microstructure and physical properties of SiC_f/SiC composites fabricated by the CVI/LSI/PIP hybrid process

Young-Seok Jeong[1,2] · Min-Ho Jang[3] · Jung-Won Kim[3] · Kyoon Choi[1] · Ho Gyu Yoon[2]

Received: 6 July 2023 / Revised: 26 September 2023 / Accepted: 8 October 2023 / Published online: 21 November 2023

Abstract

The mechanical properties of SiC_f/SiC composites depend on the microstructure of matrix, fiber and interphase. Therefore, the technology to control the microstructure of those components can be key to determining its application in the future. In this study, SiC_f/SiC composite with a fiber volume fraction of 30 vol% was prepared using the CVI/LSI/PIP hybrid process. The apparent density and residual silicon of the prepared composite were measured to be 2.77 g/cm^3 and 2 vol%, respectively. As a result of evaluatingthe tensile strength, the elastic modulus, PLS, and UTS of the SiC_f/SiC composite at 1400 °C were measured to be 152.3 GPa, 125.1 MPa, and 144.1 MPa, respectively. Additionally, the crystallinity of the PyC interphase in SiC_f/SiC composites was quantified using HRTEM, and the effect of the microstructure of the SiC matrix and PyC interphase on the mechanical properties discussed.

Keywords SiC_f/SiC composite · PyC interphase · CVI/LSI/PIP hybrid process · High-temperature strength

1 Introduction

As an effort to overcome the temperature limit of superalloys, ceramics, especially silicon carbide-based composites, have been widely studied [1, 2]. In addition to SiC with excellent specific strength, thermal conductivity and oxidation resistance, SiC fiber-reinforced SiC composites (SiC_f/SiC) show excellent ablation resistance and toughness up to 1500 °C [2]. Therefore, the composites can improve engine efficiency by raising the inner temperature of gas turbine and greatly reducing the weight of the engine. Due to these properties, the SiC_f/SiC composites became key aircraft structural materials such as nozzles, shrouds, turbine exhaust lanes, and combustors of gas turbine engines [3, 4].

To apply SiC_f/SiC composites to mechanical parts of high reliability, it is necessary to understand the failure behaviors as crack initiation and propagation in the composites. Cracks can be generated by large pore in the matrix, contraction during the matrix formation or debonding at the interfaces between the fiber and the matrix [5]. Under stress, cracks propagate through the matrix and reach the interfaces between the fiber and the matrix. The crack deflection, fiber bridging and fiber full-out suppress further crack propagation, which result in energy absorption. That is, SiC_f/SiC composites improve fracture toughness by various toughening mechanisms [6]. On the other hand, when the stress applied to the SiC_f/SiC composite reaches a critical value, the fibers break, which results in a catastrophic failure and the stress is called an ultimate tensile strength (UTS). The UTS is affected by not only the bulk properties of SiC fiber strength and fiber volume fraction, but also the interphase properties like interlayer thickness, its composition, and the interfacial shear strength [7]. The fiber bundle arrangement, bulk density, porosity, and homogeneity are other important factors for the high strength [5–7]. Since both the strength and the toughness are determined by the microstructure and the distribution of constituents, it is necessary to precisely control the microstructure between the constituents to obtain high-performance SiC_f/SiC composites.

Chemical vapor infiltration (CVI), polymer impregnation and pyrolysis (PIP), and liquid silicon infiltration (LSI) methods are mainly used for the preparation of SiC_f/SiC composites. CVI can form a high-purity SiC matrix without

✉ Kyoon Choi
knchoi@kicet.re.kr

[1] Icheon Branch, Korea Institute of Ceramic Engineering and Technology, Icheon 17303, Korea

[2] Department of Material Science and Engineering, Korea University, Seoul 02841, Korea

[3] Orange E&C, Gumi 39171, Korea

damage to the SiC fiber, but has the disadvantages of a long process time and high cost. In contrast, LSI can make a high-density matrix of low residual porosity with low cost, but since the process is performed at over 1500 °C, SiC fibers can receive some damages [8, 9]. On the other hand, PIP can proceed with the infiltration process at a relatively low temperature and has advantages of near-net-shape casting and mass production, but has poor mechanical properties due to the presence of micropores in the matrix [10]. Therefore, to compensate for the disadvantages of each process, research is being conducted to prepare an SiC_f/SiC composite by mixing two or more infiltration processes.

In this study, the interphase was formed on the SiC fiber based on the optimized pyrolytic carbon (PyC) process conditions [11], and a high-density SiC_f/SiC composite was fabricated using the CVI/LSI/PIP hybrid process. After the preparation of the SiC_f/SiC composites, the microstructural and phase analysis were done with SEM and TEM with special emphasis on the fiber and matrix interfaces. The tensile strength of the SiC_f/SiC composite was evaluated at 1400 °C.

2 Experimental

Tyranno SA3 fabric sheets (Ube Industries, LTD., Japan, W 110 mm X D 220 mm) woven alternately at 0° and 90° were used as reinforcement for the SiC_f/SiC composite (Fig. 1). The PyC was coated with a vertical CVD reactor [11] equipped with an exhaust system composed of a capacitance diaphragm gauge (CDG) and a throttle valve to automatically maintain pressure. The condition of PyC deposition was done at 10 torr and 1100 °C for 38 h flowing 600 sccm of hydrogen gas containing 5% methane through a mass flow controller (MFC) [12]. As a result, an interphase layer of 130 ± 14 nm thickness was coated on the fiber. The 25 fabrics were stacked into a graphite holder and then infiltrated with a temperature-gradient CVI (TG-CVI) system to form the SiC matrix under the previous selected condition [13, 14].

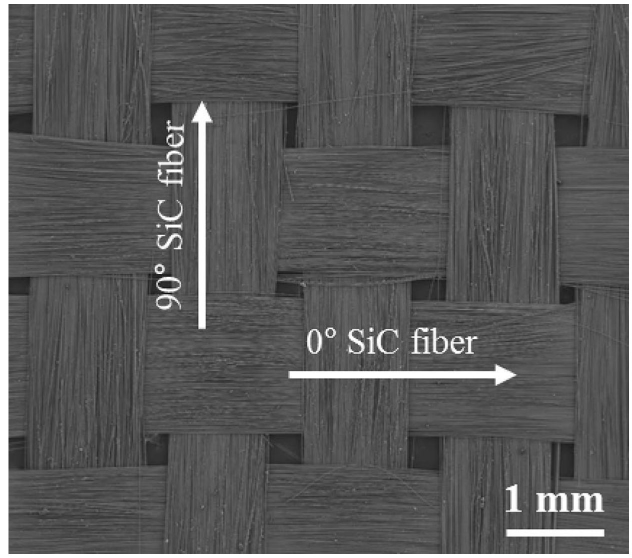

Fig. 1 SiC fabric woven with Tyranno SA3 fiber bundles of 0° and 90° orientations

After the CVI process, the preform was taken out of the graphite holder and the LSI and subsequent PIP processes were performed to fill out the remaining open pores. Firstly, a slurry for the LSI process was prepared by mixing carbon powder (particle size: 2–4 µm, Orange E&C Co. Ltd., Korea), ethyl-alcohol (99%, Dae-jung Chemical Co., Ltd., Korea), polydimethylsilane ($[(CH_3)_2Si]_n$, Dae-jung Chemical Co., Ltd., Korea), xylene ($C_6H_4(CH_3)_2$, 99%, Dae-jung Chemical Co., Ltd., Korea), and methyl ethyl ketone (C_4H_8O, 98%, Dae-jung Chemical Co., Ltd., Korea) as a dispersing agent to suppress aggregation between carbon particles. The slurry was impregnated into the preform in a vacuum, dried at 60 °C, and then heat treated at 600 °C in N_2 for 1 h. SiC was formed at 1450 °C in a vacuum by supplying Si powder, where the weight of the supplied Si powder was 70% of the weight of the impregnated carbon powder to minimize residual Si in the SiC_f/SiC composite. Next, the PIP process adopted the method suggested by Koo et al. [15], and after impregnating the preform using

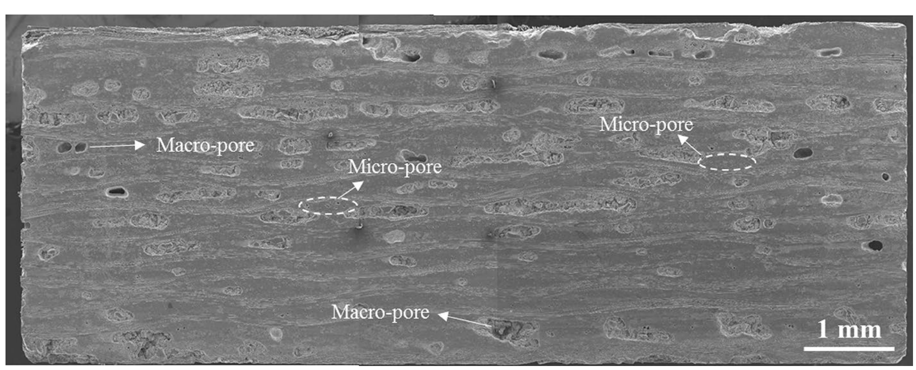

Fig. 2 Overall cross-sectional micrographs of SiC_f/SiC specimen fabricated by CVI/LSI/PIP hybrid process

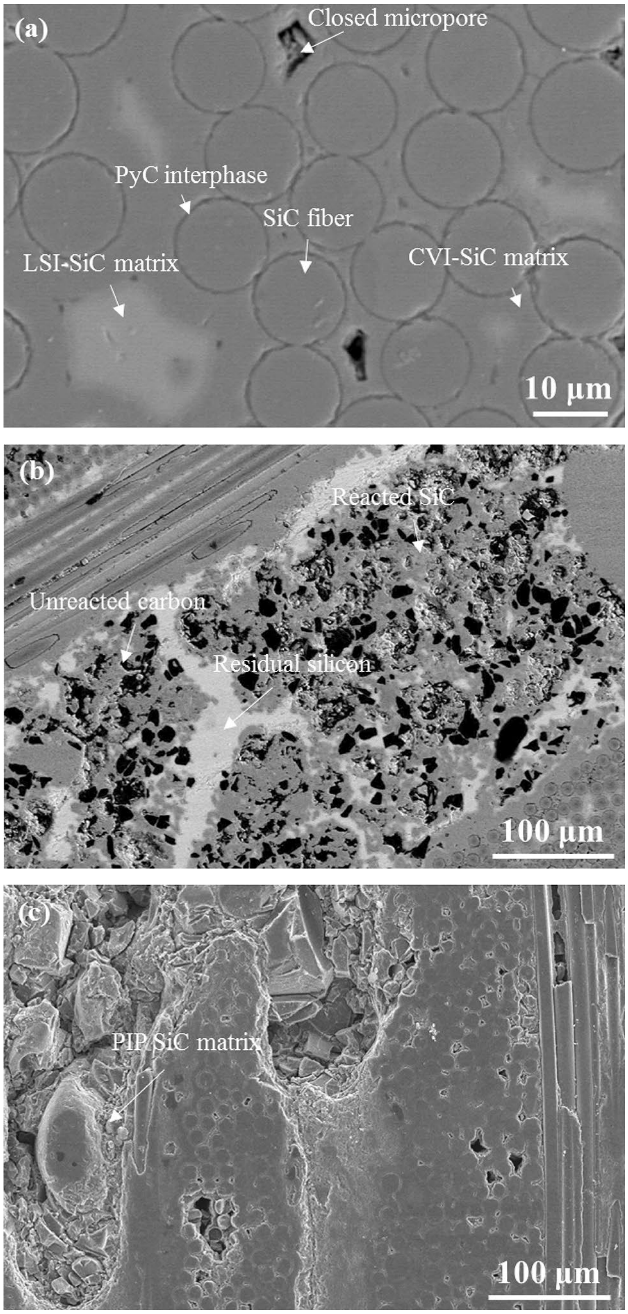

Fig. 3 SEM micrographs of SiC_f/SiC composites in local magnification showing **a** CVI–SiC matrix around SiC fiber, **b** LSI-processed matrix near SiC bundle and **c** PIP-filled SiC matrix between SiC bundles

the polycarbosilane (PCS, SMP-10, Starfire Systems, Inc., USA), heat treatment was performed at 1300 °C in a vacuum. The PCS impregnation and heat treatment process were repeated 6 times to prepare an SiC_f/SiC composite.

The final apparent density of the SiC_f/SiC composite prepared by the hybrid process is 2.77 g/cm^3, and the amount of residual Si in the SiC matrix is less than 2 vol%. The microstructure and fracture surfaces of the SiC_f/SiC composite were observed with a scanning electron microscope (JSM-6701F, JEOL Co., Ltd., Japan). Then, the interphase in the SiC_f/SiC composite was analyzed using a transmission electron microscope (ARM200, JEOL Co., Ltd., Japan).

A high-temperature tensile strength test was performed to confirm the mechanical properties of the SiC_f/SiC composite. Strength specimens were cut into dog-bone-shaped ($152 \times 4 \times 8$ mm^3) specimens according to the ASTM C 1359-11 standard as suggested by Kim et al. [16]. The tensile strength was measured with an ultimate test system (370 Load Frame, MTS systems Co., Ltd.) at 1400° C in air. Load cell of 500 kN capacity was used and loading speed was 0.5 mm/min and the length of the strain gauge was 12.5 mm. To observe the fracture surface, the specimen was air-quenched to room temperature immediately after fracture.

3 Results and discussion

3.1 Microstructure of hybrid processed SiC_f/SiC composite

Figure 2 shows a cross section of the SiC_f/SiC specimen fabricated by the hybrid process in which 0° fiber bundles and 90° fiber bundles are crossed. SiC matrix was formed inside the fiber bundles (intra-bundle) and between the fiber bundles (inter-bundle). Macropores with a size of 100 μm or more were left between the fiber bundles, and micropores with a size of a few to several micrometers were formed inside the fiber bundle. As such, the distribution of micropores showed a bimodal distribution pattern in which macropores between fiber bundles and micropores within the bundles coexisted, which is closely related to the infiltration process. The various microstructures were observed depending on the process used, which is shown in Fig. 3.

First, in the CVI process, deposition started from the fiber surface to form a dense SiC matrix. In addition, it can be confirmed that closed micropores are formed in the fiber bundle in the length direction of the fiber (Fig. 3a). It is assumed that SiC is formed on the fiber surface and as the layer becomes thicker, it blocks the pore channels through which gas penetrates, forming closed micropores in the fiber bundle [17]. During this process, it is thought that long and narrow micropores are formed along the fiber axis in the intra-bundle with a narrow spacing between the fibers, and the effect of these pores on the mechanical properties is considered to be relatively insignificant.

Empty space between fiber bundles after the CVI process leave large pore channels which can be filled by the subsequent LSI process. The LSI–SiC matrix thus formed consists of the reacted SiC phase and residual Si and C

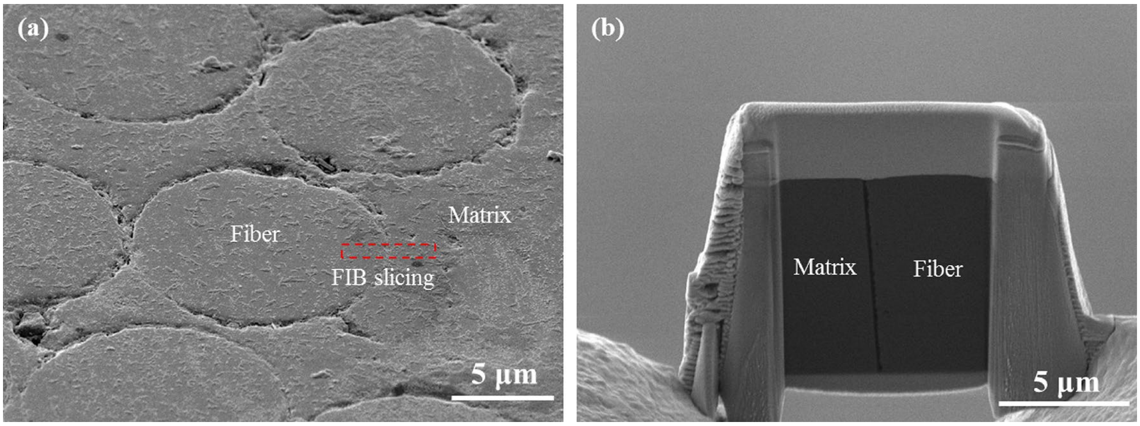

Fig. 4 **a** SEM cross-sectional image of SiC_f/SiC composite with the FIB-sliced region (dotted rectangle) and **b** the extracted thin foil

(Fig. 3b). In the LSI process, molten silicon first reacts with a carbon source at the surface to form SiC and diffuses inward through the reacted SiC, but the reacted SiC hinders the diffusion of silicon into the carbon source, remaining as unreacted carbon [18]. As the LSI process temperature increases, the fraction of residual Si and carbon decreases. However, since the infiltration process at high temperatures causes damage to the SiC fibers, the temperature of the LSI process was lowered to about 1450 °C. The fraction of the residual Si in the LSI–SiC matrix was adjusted to less than 2% by supplying 70% Si powder to the weight of the impregnated carbon powder.

Even after the CVI and LSI processes were completed, large pores remained between the bundles. These open pores were filled with SiC by a repeated PIP process, and it easily infiltrated the inside of the preform through the open pore channel, forming a PIP–SiC matrix (Fig. 3c). SiC formed in this process forms microcrystalline or amorphous silicon carbide [19], and large volumetric shrinkage occurs during heat treatment, leaving residual pores and microcracks [20].

3.2 HRTEM analysis on the PyC interphase and its domain structure

The PyC interphase between the fiber and the matrix serves to improve the toughness of the SiC_f/SiC composite by inducing crack deflection and debonding. The toughness of the SiC_f/SiC composite depends on the microstructure of the PyC interphase. A specimen prepared with a focused ion beam (FIB) system was prepared to observe the microstructure of the PyC interphase in the SiC_f/SiC composite as shown in Fig. 4a. The thinly sliced specimen (Fig. 4b) was attached to a TEM grid and the microstructure and crystallinity of the interphase were analyzed with an HRTEM.

The PyC interphase of the SiC_f/SiC composite is mostly formed in a turbostratic structure which is an in-between form, a mixture of sp^2 and sp^3 hybridized carbon: graphitic, sp^2-hybridized domains are surrounded by an amorphous matrix [18]. It is accompanied by an increase in the interplanar spacing [11, 22, 23]. These defective graphene layers constitute nanodomains and are randomly stacked in the PyC interphase. Depending on the arrangement of the graphene layers, a diffraction point or an arc-shaped azimuthal intensity distribution occurs in the reciprocal space. In highly crystalline structures such as highly oriented pyrolytic graphite (HOPG), diffraction points appear in reciprocal space, whereas turbostratic structures appear as azimuthal intensity distributions in reciprocal space due to many defects. Figure 5 shows the HRTEM image and the corresponding fast Fourier transform (FFT) image depending on the arrangement of nanodomains in the PyC interphase. Diffraction points appear in the structure in which graphene layers are stacked in a certain direction and spacing, which means that nanodomains with high crystallinity are formed (Fig. 5a). However, when the PyC layer is mixed with curved graphene sheets, increased spacing between graphene layers, and distortion defects, it appears as an azimuthal intensity distribution in the FFT (Fig. 5b). The width of the azimuthal intensity distribution gradually widens as the number of structural defects in the PyC layer increases (Fig. 5c). That is, the width of the azimuthal intensity distribution can be used as a criterion for evaluating the degree of disorder of the graphene layers. The crystallinity of the PyC interphase can be quantified by comparing them.

The crystallinity of the PyC interphase can be quantified by the orientation angle (OA) calculated as the width of the orientation intensity distribution of the (002) lattice plane in the reciprocal lattice space. Turbostratic structures are classified into high texture (HT, $OA \le 50°$), medium texture (MT, $80° \ge OA \ge 50°$), and low texture (LT, $180° \ge OA \ge 80°$) depending on the orientation angle [24]. To extract the

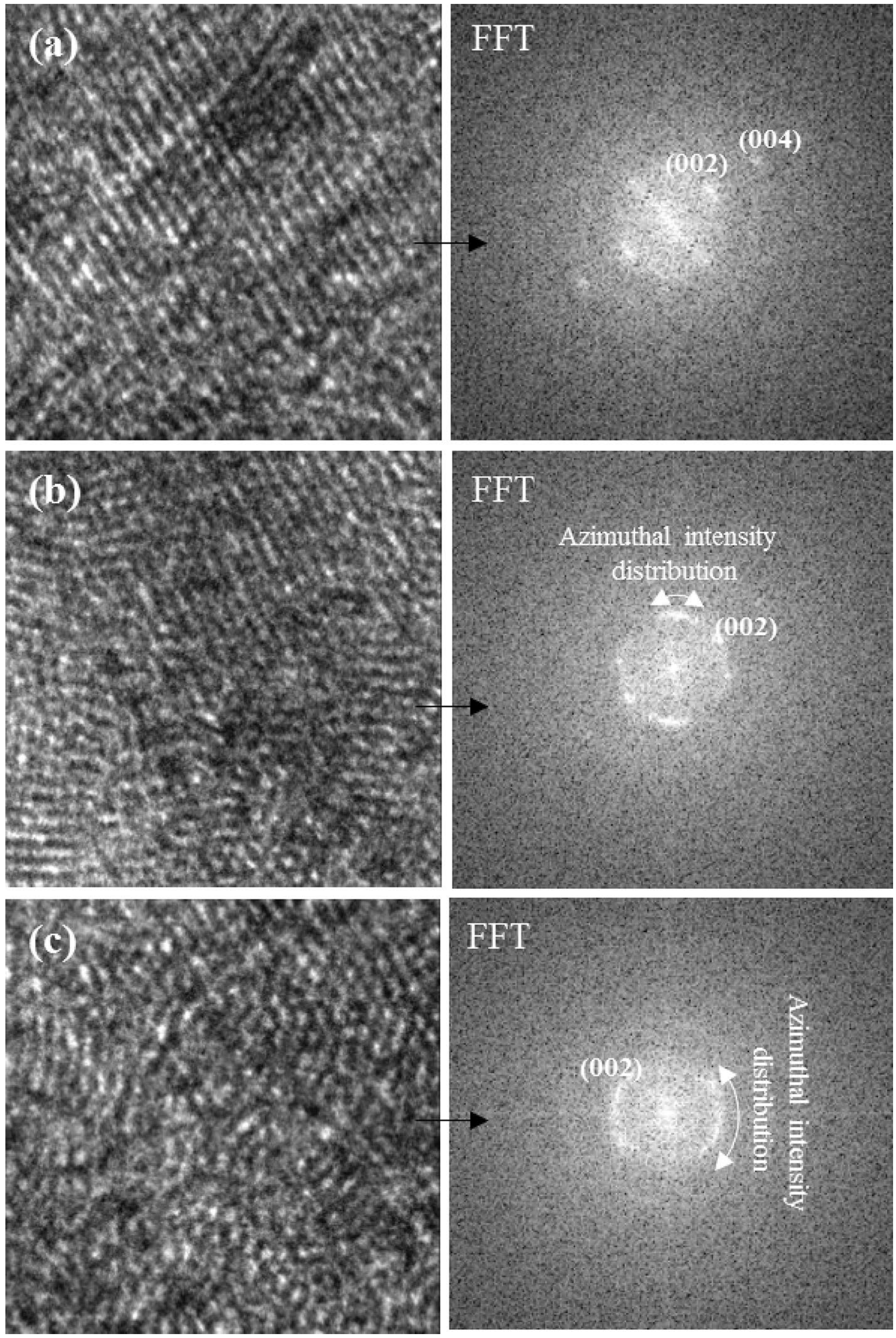

Fig. 5 HRTEM (left) and its corresponding FFT (right) images of **a** PyC stacked with highly oriented nanodomains, **b** PyC stacked with a mixture of highly oriented nanodomains and randomly oriented ones, and **c** PyC stacked with randomly oriented nanodomains

orientation angle, the calculation method proposed by Seyring [25] was adopted, and GATAN's DigitalMicrograph software (DM3) was used as the analysis software [26]. The diffraction pattern images were obtained from the SAED pattern by randomly selected five regions of PyC between the SiC fiber and SiC matrix (Fig. 6). The average OA of the PyC interphase obtained from the diffraction pattern image was calculated to be 47.5° (±6.1). Among the selected areas, four areas were found to have OAs corresponding to HT PyC and one had an orientation angle corresponding to MT PyC where the OA of 58.9° was measured. According to J. Kabel et al. [27], it was confirmed to have similar physical properties within ~10° of transition from the HT to MT structure.

Within a turbostratic structure, the nanodomain region is not perfectly flat like graphite, but has a structure in which graphene sheets are arranged at regular intervals. Digital image processing is widely used to analyze the length (L_a) and interplanar spacing (d_{002}) of graphene sheets in nanodomains in turbostratic structures such as coal, soot, and amorphous carbon by removing information on stacking faults or

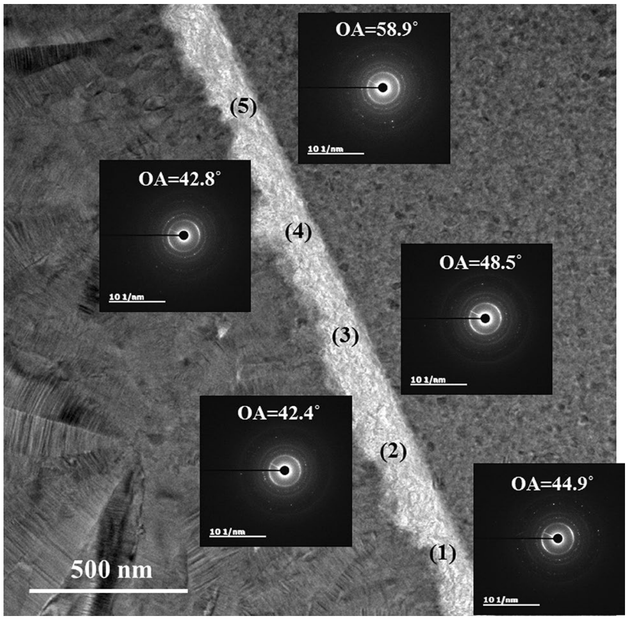

Fig. 6 TEM image between the SiC fiber and CVI–SiC and the SAED patterns of 5 distinct regions along the PyC interphase with orientation angle (OA) noted

amorphous structures from HRTEM images [28–30]. This method can extract L_a and d_{002} in the nanodomain after simplifying the lattice fringes with complex shapes through the process of standardization, binarization, and skeletonization of the lattice fringe image obtained from HRTEM. Recently, P. Toth [21] developed an algorithm with high reliability and identified L_a, d_{002} in the nanodomain of turbostratic carbon material. This study also utilized this method to extract L_a, d_{002} in the nanodomain in the PyC interphase.

During digital image processing, the pixel size was fixed at 0.023 nm/pixel. By randomly selecting 5 regions, L_a and d_{002} of PyC were collected. The collected values were standardized by the length calculated from the HRTEM image (X-axis) and the number of measured data (Y-axis) as shown in Fig. 7. L_a in PyC was calculated to be 0.64 nm on average, which means that two or three hexagonal ring lattices with a length of $d_a = 0.246$ nm are connected to form the graphene layer. d_{002} was calculated to be 0.345 nm, which appears to be slightly higher than that of the graphite crystal (0.334 nm) due to stacking faults [23]. These changes in L_a and d_{002} may affect the mechanical properties of the SiC_f/SiC composite.

Cracks created in the matrix are deflected by the strongly bonded graphene layer in the PyC interphase and absorb energy by separating the van der Waals bonded graphene layers [31]. If d_{002} increases due to point defects and stacking faults, debonding easily occurs in the PyC layer because the bonding force between the graphene layers is very weak. This makes the load transfer from the matrix

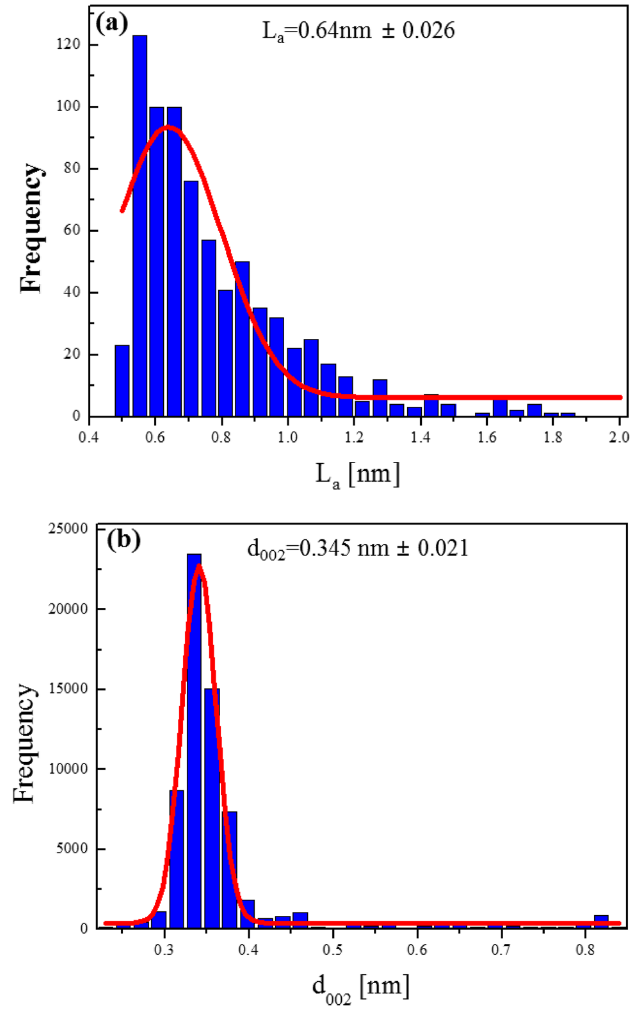

Fig. 7 Standard normal distribution of lattice parameters **a** L_a and **b** d_{002} of PyC interphase in the SiC_f/SiC composites

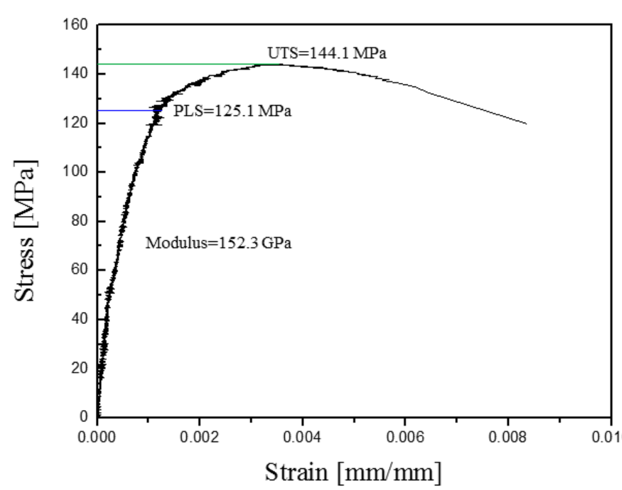

Fig. 8 Stress–strain curve of the SiC_f/SiC composite fabricated by the CVI/LSI/PIP hybrid process at 1400 ℃ in air

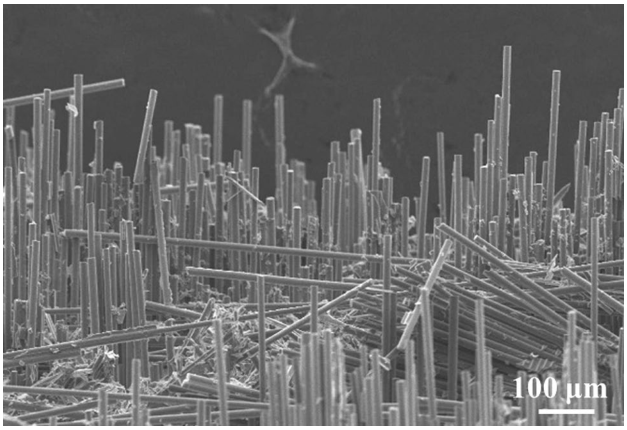

Fig. 9 Fiber pullout of the SiC_f/SiC composite fabricated by the CVI/LSI/PIP hybrid process

to the fibers difficult and reduces the strength of the SiC_f/SiC composite. On the other hand, highly oriented PyC can adequately maintain the interfacial bonding strength between the matrix and the fibers, allowing sufficient load transfer from the matrix to the fibers. As a result, the strength of the SiC_f/SiC composite are improved. The fractured fibers then slide along the debonded interface and absorb crack energy. The energy absorbed during fiber pullout depends on the interfacial shear strength between the fiber and matrix [32, 33], where the length of the PyC graphene sheet affects the interfacial shear strength. According to Zhou et al. [34], small-sized graphene sheets that are easily deformed form strong bonds as they meet each other under shear load, which increases the interfacial shear stress and can dissipate a lot of energy. Therefore, PyC formed with small graphene lengths in the SiC_f/SiC composite seems to be helpful in improving the toughness of the SiC_f/SiC composite.

3.3 Measurement of the tensile strength of SiC_f/SiC composites at high temperature

The stress–strain curve of the SiC_f/SiC composite at 1400 °C in air was shown in Fig. 8. The fracture behavior of the SiC_f/SiC composite can be divided into three regions. The first region is the elastic deformation and the modulus of the SiC_f/SiC composite was measured to be 152.3 GPa. The premature cracks are formed from the inherent defects like pores or cracks in the matrix, but do not propagate throughout the composite [35]. Elastic deformation appears up to the proportional limit strength (PLS) at which the crack propagation is initiated [36]. Further stress loading over the PLS of the composite induces the crack propagation through the matrix. During the stress loading to the UTS, the strain through the toughening mechanisms such as crack deflection, bridging, and debonding at the interphase resulted in a nonlinear stress–strain curve [37]. The UTS depending on the SiC fiber properties such as fiber strength, volume fraction, architecture, and defects [38] reached 144.1 MPa in this study. After the maximum is reached, the fiber is broken, and energy is dissipated during the sliding of the failure fiber along the debonded interface. As the interfacial shear stress increases, the amount of energy dissipation increases [39]. The pullout SiC fiber can be observed on the fracture surface as shown in Fig. 9. This kind of fiber pullout behavior is achieved by a combination of the ideal interfacial properties, sufficient density, and low damage of the SiC fibers through the low heat treatment temperature. Therefore, it is not a frequently observed behavior and could only be observed in the special samples.

Since the SiC_f/SiC composite manufactured by CVI is deposited on fibers as a substrate, the reactive gas cannot easily reach the area where the gap between fibers is narrow [40]. The deposition rate gets slower as the densification proceeds. And as a result, the large empty space between fiber bundles is hard to fill out and remained after the CVI process. Therefore, an additional process to effectively fill the empty space is required. Depending on the final purpose of use, PIP or LSI is used to fill the remaining pores with the SiC matrix. The mechanical properties of the SiC_f/SiC composite prepared by the hybrid process have a great influence depending on the additional process. Table 1 compares the physical properties of SiC_f/SiC composites prepared with CVI/PIP, CVI/LSI and CVI/LSI/PIP. In the case of the composite (N24-C) manufactured by CVI/LSI, it has excellent properties even at 1300 °C, but the mechanical properties may deteriorate at 1400 °C. This is because the remaining silicon attacked the fibers and interphases when maintained at 1400 °C in an air atmosphere. On the other hand, the composite made of CVI/PIP (N26-A) maintained

Table 1 Comparison of physical properties of SiC_f/SiC composites prepared by the various hybrid processes

	SiC infiltration process	SiC fiber (vol%)	Density (g/cm^3)	PLS (test temperature)	Residual Si (vol%)	References
N24-C	CVI + LSI	36	2.76	150 MPa (1300℃)	18	[41]
N26-A	CVI + PIP	36	2.52	120 MPa (1400℃)	0	[41]
This study	CVI + LSI + PIP	30	2.77	125.1 MPa (1400℃)	< 2	–

higher mechanical strength than N24-C at 1400 °C because residual silicon did not exist in the matrix [41].

The physical properties of the three composites were compared in Table 1. Direct comparison may be difficult for the three composites because they differ in the SiC fiber, volume fraction of each phase, the interface structure and so on [36, 41]. The SiC_f/SiC composite prepared by CVI/LSI/PIP showed a 5% increase in PLS at 1400 °C, while the fiber volume fraction was lower than that of N24-C and N26-C. Therefore, it was possible to improve the mechanical properties of the SiC_f/SiC composite at high temperature by forming an optimized SiC matrix through the CVI/LSI/PIP process. Although about 2 v/o silicon remains in the SiC matrix of the SiC_f/SiC composite manufactured by CVI/LSI/PIP, it is judged to have excellent physical properties at 1400 °C because the SiC/PyC layer is well protected by the thick CVI–SiC layer. The hybrid process presented here minimizes the CVI process, which is the most expensive process, while minimizing porosity and reducing residual silicon to 2 v/o. When SiC_f/SiC composites are used for long-term use at high temperatures above 1400 °C, this hybrid process is a useful method. And additional researches are needed to lower the unit cost of SiC_f/SiC components.

4 Conclusion

SiC_f/SiC composites were successfully fabricated by the CVI/LSI/PIP hybrid process. The CVI process densely wraps around the fiber and forms a matrix to make a preform with some strength. In the next LSI process, the intra-bundle pores remained after the CVI process was completed. The larger pores between bundles were filled to make the overall final density up to 98% after the final PIP process. A highly oriented PyC interphase was formed between the fiber and the matrix using hydrogen gas containing CH_4 5%. The PyC interphase of high textured (HT) turbostratic structure having nanodomains was analyzed with HRTEM images and digital processing. As a result, the PLS and UTS of the SiC_f/SiC composite at 1400 °C in air reached 125.1 MPa and 144.1 MPa, respectively.

Declarations

Conflict of interest The authors declare no conflict of interest.

References

1. P. Wang, F. Liu, H. Wang, H. Li, Y. Gou, A review of third generation SiC fibers and SiC_f/SiC composites. J. mater. sci. technol. **35**, 2743–2750 (2019). https://doi.org/10.1016/j.jmst.2019.07.020
2. D. Wei, L. Lei, J. Yinghou, W. Sontao, L. Xingchen, B. Sunden, Heat transfer in the trailing region of gas turbines – A state-of-the-art review. Appl. Therm. Eng. **199**, 117614 (2021). https://doi.org/10.1016/j.applthermaleng.2021.117614
3. M. Chen, H. Qiu, W. Xie, B. Zhang, S. Liu, W. Luo, X. Ma, Research progress of continuous fiber reinforced ceramic matrix composite in hot section components of aero engine. IOP Conf Ser Mater Sci Eng **678**, 012043 (2019). https://doi.org/10.1088/1757-899X/678/1/012043
4. P.G. Karandikar, T.-W. Chou, Damage Development and Moduli Reductions in Nicalon-Calcium Aluminosilicate Composites under Static Fatigue and Cyclic Fatigue. J. Am. Ceram. **76**, 1720–1728 (1993). https://doi.org/10.1111/j.1151-2916.1993.tb06640.x
5. A.G. Evans, F.W. Zok, The physics and mechanics of fibre-reinforced brittle matrix composites. J. Mater. Sci. **29**, 3857–3896 (1994). https://doi.org/10.1007/BF00355946
6. J.D. Arregui-Mena, T. Koyanagi, E. Cakmak, C.M. Petrie, W.-J. Kim, D.J. Kim, C.P. Deck, C. Sauder, J. Braun, Y. Katoh, Qualitative and quantitative analysis of neutron irradiation effects in SiC/SiC composites using X-ray computed tomography. Compos. B Eng. **238**, 109896 (2022). https://doi.org/10.1016/j.compositesb.2022.109896
7. W. Yang, T. Noda, H. Araki, J. Yu, A. Kohyama, Mechanical properties of several advanced Tyranno-SA fiber-reinforced CVI–SiC matrix composites. Mater. Sci. Eng. C **345**, 28–35 (2003). https://doi.org/10.1016/S0921-5093(02)00468-9
8. S.Y. Kim, I.S. Han, S.K. Woo, K.S. Lee, D.K. Kim, Wear-mechanical properties of filler-added liquid silicon infiltration C/C–SiC composites. Mater. Des. **44**, 107–113 (2013). https://doi.org/10.1016/j.matdes.2012.07.064
9. J. W. Han, D. J. Kim, S.-Y. KIM, W.-J. Kim, C. Park, J. Y. Park, Influence of CVI process time on the C-ring strength of hybrid SiC_f/SiC composites fabricated by CVI–LSI. J Korean Ceram. **60** 261–271 (2023). https://doi.org/10.1007/s43207-022-00260-7
10. K.-M. Kim, Y.S. Hahn, S.-M. Lee, K. Choi, J.-H. Lee, Mechanical properties of Cf/SiC composite using a combined process of chemical vapor infiltration and precursor infiltration pyrolysis. J. Korean Ceram. Soc. **55**, 392–399 (2018). https://doi.org/10.4191/kcers.2018.55.4.11
11. Y.S. Jeong, K. Choi, H.G. Yoon, Microstructural control of pyrolytic carbon layer deposited from methane by isotropic chemical vapor infiltration. J. Korean Ceram. Soc. **56**, 291–297 (2019). https://doi.org/10.4191/kcers.2019.56.3.09
12. N. Orlovskaya, M. Lugovy, F. Ko, S. Yarmolenko, J. Sankar, J. Kuebler, SiC/SiC_{wf} Laminates: design, manufacturing, mechanical PROPERTIES. Compos. B Eng. **37**, 524–529 (2006). https://doi.org/10.1016/j.compositesb.2006.02.022
13. K.M. Kim, J.-W. Seo, K. Choi, Improvement of densification uniformity of carbon/silicon carbide composites during chemical vapour infiltration. Int. J. Nanotechnol. **15**, 555–567 (2018). https://doi.org/10.1504/IJNT.2018.096346
14. J.-W. Seo, K. Choi, Application of CFD simulation to silicon carbide deposition for nozzles with funnel. J. Korean Ceram. **58**, 184–191 (2021). https://doi.org/10.1007/s43207-020-00082-5
15. J.-M. Koo, K.H. Kim, Y.S. Han, Fabrication and characterization of Cf/SiC composite with bn interphase coated by wet chemical process. J. Korean Inst. Surf. Eng. **50**, 523–530 (2017). https://doi.org/10.5695/JKISE.2017.50.6.523
16. K. H. Kim, K. Choi, Y. S. Han, S. Nahm, S.-M. Lee, High-temperature cyclic fatigue in air of SiC_f/SiC ceramic matrix composite with a pyrolytic carbon interface. HTHP. **50,** 325–334 (2021). https://doi.org/10.32908/hthp.v50.1139
17. R. Liu, F. Wang, J. Zhang, J. Chen, F. Wan, Y. Wang, Effects of CVI SiC amount and deposition rates on properties of SiC_f/SiC

composites fabricated by hybrid chemical vapor infiltration (CVI) and precursor infiltration and pyrolysis (PIP) routes. Ceram. Int. **47**, 26971–26977 (2021). https://doi.org/10.1016/j.ceramint.2021.06.110
18. J.C. Margiotta, D. Zhang, D.C. Nagle, C.E. Feeser, Formation of dense silicon carbide by liquid silicon infiltration of carbon with engineered structure. J. Mater. Res. **23**, 1237–1248 (2008). https://doi.org/10.1557/JMR.2008.0167
19. Y.J. Joo, S.H. Joo, C.K. Youn, Effect of pyrolysis temperature and pressing load on the densification of amorphous silicon carbide block. J. Korean Cryst. Growth Cryst. Technol. **30**, 271–276 (2020). https://doi.org/10.6111/JKCGCT.2020.30.6.271
20. T.J. Pirzada, S. Singh, R.D. Meyere, P. Earp, M. Galano, T.J. Marrow, Effects of polymer infiltration processing (PIP) temperature on the mechanical and thermal properties of Nextel 312 fibre SiCO ceramic matrix composites. Compos. Part A Appl. **140**, 106197 (2021). https://doi.org/10.1016/j.compositesa.2020.106197
21. P. Toth, Nanostructure quantification of turbostratic carbon by HRTEM image analysis: State of the art, biases, sensitivity and best practices. Carbon **178**, 688–707 (2021). https://doi.org/10.1016/j.carbon.2021.03.043
22. P. Ruz, S. Banerjee, M. Pandey, V. Sudarsan, P.U. Sastry, R.J. Kshirsagar, Structural evolution of turbostratic carbon: Implications in H_2 storage. Solid State Sci. **62**, 105–111 (2016). https://doi.org/10.1016/j.solidstatesciences.2016.10.017
23. R. Liu, Y. Chi, L. Fang, Z. Tang, Z. Tang, Synthesis of Carbon Nanowall by Plasma-Enhanced Chemical Vapor Deposition Method. J. Nanosci. Nanotechnol. **14**, 1647–1657 (2014). https://doi.org/10.1166/jnn.2014.8905
24. P. J. Meadows, E. L. Honorato, P. Xiao, Fluidized Bed Chemical Vapor Deposition of Pyrolytic Carbon - II. Effect of Deposition Conditions on Anisotropy. Carbon **47,** 251–62 (2009). https://doi.org/10.1016/j.carbon.2008.10.003
25. M. Seyring, A. Simon, I. Voigt, U. Ritter, M. Rettenmayr, Quantitative crystallographic analysis of individual carbon nanofibers using high resolution transmission electron microscopy and electron diffraction. Carbon **116**, 347–355 (2017). https://doi.org/10.1016/j.carbon.2017.01.107
26. K. Kamei, T. Shimizu, K. Harano, E. Nakamura, Aryl radical addition to curvatures of carbon nanohorns for single-molecule-level molecular imaging. Bull. Chem. Soc. Jpn **93**, 1603–1608 (2020). https://doi.org/10.1246/bcsj.20200232
27. J. Kabel, T.E.J. Edwards, A. Sharama, J. Michler, P. Hosemann, Direct observation of the elasticity-texture relationship in pyrolytic carbon via in situ micropillar compression and digital image correlation. Carbon **182**, 571–584 (2021). https://doi.org/10.1016/j.carbon.2021.06.045
28. P. Toth, A.B. Palotas, E.G. Eddings, R.T. Whitaker, J.S. Lighty, A novel framework for the quantitative analysis of high resolution transmission electron micrographs of soot I. Improved measurement of interlayer spacing. Combust. Flame **160**, 909–919 (2013). https://doi.org/10.1016/j.combustflame.2013.01.002
29. K. Yehliu, R. L. Vander Wal, A. L. Boehman, Development of an HRTEM image analysis method to quantify carbon nanostructure. Combust. Flame **158,** 1837–1851 (2011). https://doi.org/10.1016/j.combustflame.2011.01.009
30. C.A. Wang, T. Huddle, E.H. Lester, J.P. Mathews, Quantifying curvature in high-resolution transmission electron microscopy lattice fringe micrographs of coals. Energy Fuels **30**, 2694–2704 (2016). https://doi.org/10.1021/acs.energyfuels.5b02907
31. J.M. Parente, P. Santos, S. Valvez, M.P. Silva, P.N.B. Reis, Fatigue behaviour of graphene composites: an overview. Procedia Struct. Integr. **5**, 282–293 (2020). https://doi.org/10.1016/j.prostr.2020.04.033
32. C. Droillard, J. Lamon, Fracture toughness of 2-D Woven SiC/SiC CVI-composites with multilayered interphases. J. Am. Ceram. **79**, 849–858 (1996). https://doi.org/10.1111/j.1151-2916.1996.tb08516.x
33. F. Rebillat, J. Lamon, The concept of a strong interface applied to SiC/SiC composites with a BN interphase. Acta Mater. **48**, 4609–4618 (2000). https://doi.org/10.1016/S1359-6454(00)00247-0
34. Y. Zhou, T. Ye, L. Ma, Z. Lu, Z. Yang, S. Liu, Investigation on Cf/PyC interfacial properties of C/C composites by the molecular dynamics simulation method. Mater. **25**, 679 (2019). https://doi.org/10.3390/ma12040679
35. R.K. Goldberg, A.S. Almansour, R.M. Sullivan, Analytical simulation of effects of local mechanisms on tensile response of ceramic matrix minicomposites. J. Eur. Ceram. **42**, 6846–6864 (2022). https://doi.org/10.1016/j.jeurceramsoc.2022.07.042
36. L. Li, Temperature-dependent proportional limit stress of SiC/SiC fiber-reinforced ceramic-matrix composites. High Temp. Mater. Process. **39**, 0052 (2020). https://doi.org/10.1515/htmp-2020-0052
37. A. Haboub, H.A. Bale, J.R. Nasiatka, B.N. Cox, D.B. Marshall, R.O. Ritchie, A.A. MacDowell, Tensile testing of materials at high temperatures above 1700 °C with in-situ synchrotron X-ray micro-tomography. Rev. Sci. Instrum. **85**, 083702 (2014). https://doi.org/10.1063/1.4892437
38. K. Shimoda, T. Hinoki, Effects of fiber volume fraction on the densification and mechanical properties of unidirectional SiCf/SiC-matrix composites. J. Eur. Ceram. **41**, 1163–1170 (2021). https://doi.org/10.1016/j.jeurceramsoc.2020.09.024
39. H.A. Bale, A. Haboub, A.A. MacDowell, J.R. Nasiatka, D.Y. Parkinson, B.N. Cox, D.B. Marshall, R.O. Ritchie, Real-time quantitative imaging of failure events in materials under load at temperatures above 1,600°C. Nat. Mater. **12**, 40–46 (2013). https://doi.org/10.1038/nmat3497
40. J.Y. Park, D.J. Kim, H.G. Lee, W.-J. Kim, Fabrication and evaluation of C-ring strength of SiC_f/SiC composites tube. J. Korean Ceram. **58**, 718–727 (2021). https://doi.org/10.1007/s43207-021-00146-0
41. J. A. Di Carlo, H.-M. Yun, G. N. Morscher, R. T. Bhatt, SiC_f/SiC composites for 1,600°C and above. In: Bansal NP (eds) Handbook of ceramic composites. Kluwer Academic Publishers, Boston, pp 77–98 (2005). https://doi.org/10.1007/0-387-23986-3_4